

ALTIRIG · DESIGN DECISION RECORD

AltiRig TRL 2 review decisions

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/altirig		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	TRL 2 review decisions D1 to D14, decided under Amish's 2026-10-03 pre-approvals

0001: TRL 2 review decisions

- **Date:** 2026-10-03
- **Status:** decided. Amish, 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." And for this second batch, Amish, 2026-10-03: "Proceed with the remaining 15 scaffolds". Each decision below is the recommendation made at the TRL 2 review and is decided as recommended. Requirements that are not met or at risk are not decided here; they are posed to Amish in `docs/REVIEW.md` and the design decisions register.

■ Context

The scaffold (ALR-PRC-001 v0.1) described a sealed vessel with a thrust stand inside and left open the vessel type, whether to chill the air, how to deal with the propeller's recirculating wake, and whether a propeller and motor kit belongs in this repo. The TRL 2 review settled these so the TRL 3 calculations (ALR-CAL-001), the model and the build plan could proceed. Choices that touch safety take the conservative option, and each says what evidence would relax it.

Safety: AltiRig is a vacuum vessel with a spinning propeller inside, run from mains power. These decisions set the design basis on paper only. Nothing here makes the chamber safe to use; it is not certified pressure equipment or test equipment.

■ Decisions

Table 1. TRL 2 review decisions.

#	Decision	Reason	Affects
D1	Match air density, not pressure: R1's 0.74 kg/m ³ is reached at about 61.9 kPa at 20 °C; pressure and temperature are logged with every point	Thrust and power scale with density; at room temperature, 54 kPa would give the density of about 6,100 m	R1; controller; ALR-CAL-001 A
D2	Vessel: a horizontal 813 mm (32 in) x 6.35 mm steel pipe offcut, 1,500 mm long, with two bought 2:1 ellipsoidal tank heads; one head welded on the rear, one on a flange ring as the door	A thin steel drum collapses at a few kPa of vacuum; a flat-panelled box needs heavy ribs; pipe and tank heads are stocked items a welding shop can join	Vessel parts; R5, R7; cost
D3	Design the vessel for full vacuum (101.3 kPa difference) and apply R7's factor of 4 at full vacuum, not only at the working point	Conservative: the pump can reach near full vacuum if the relief valve sticks. Relax only with a code calculation by a qualified engineer and a strain-gauged proof test	R7; ALR-CAL-001 B
D4	Add R12: an adjustable vacuum relief valve opens at 55 kPa below atmosphere, so the chamber cannot go below 46.3 kPa absolute in normal use	Covers every working point (lowest 57.4 kPa) with margin and limits the load on the window and the door	R12; valve manifold
D5	Add R11: the motor can be armed only with the door latched (interlock switch) and the key turned; the emergency stop removes motor and pump power; the bleed valve is normally open, so the chamber vents on loss of power	The door is held shut by air under vacuum, so the hazard is a running propeller when the door is open or a run that cannot be stopped	R11; cabinet; build plan safety stops
D6	Window: clear polycarbonate 20 mm, never acrylic or glass, placed with its nearest edge more than 15° from the propeller plane	Polycarbonate yields rather than shatters; keeping it out of the fragment zone protects the one non-steel wall. Relax only with impact tests of the window at TRL 4	Window; R7, R11
D7	Thrust stand: two spring-steel flexure leaves (no rails), a 10 kg bending-beam cell for thrust, a torque head on two bearings with an arm on a 5 kg cell, an optical speed sensor	No sliding friction; bought C3 cells; the share carried by the leaves is fixed and calibrated out	R3, R4; stand parts
D8	Room temperature only in the first version; chilling towards -20 °C is a later variant	Density, not temperature, sets thrust; chilling a 0.9 m ³ vessel adds a refrigeration plant and condensation	Scope; R1
D9	Closed chamber with a removable 63 % open perforated baffle, and side-by-side comparison runs at ambient pressure published with every data set	The scaffold's open question on recirculation; the baffle breaks up the jet and swirl, and the comparison runs measure what the loop does. Whether R6 is met is posed to Amish separately	R6; baffle; ALR-CAL-001 F
D10	No lithium or sodium-ion battery inside the chamber in the first version; the motor is powered from a 1,500 W supply outside through the feed-through	Conservative: a battery fire in a closed vessel raises the pressure and fills it with toxic fumes. Relax only with a vent, a fire-containment plan and abuse tests of the pack type	Scope; ColdCell; feed-through
D11	Co-design: the first candidate partner to approach is the Department of Aerospace Engineering, IIT Kanpur; the second is the Department of Mechanical Engineering,		ALR-PRB-001 co-design

#	Decision	Reason	Affects
	Kathmandu University. First candidate region: the Nepal and Indian Himalaya. Recorded as candidates, not agreed	Propeller facilities and Himalayan users; a partner reviews the vessel calculation and runs the open-air comparison	
D12	The high-altitude propeller, motor and tuning kit is not part of this repo; AltiRig publishes measured data and the Kitewright Lift and Range frames publish their kits from it	Keeps AltiRig a test rig and the kits with the aircraft they serve	Scope; shared blocks
D13	Shared blocks: CalRig's proof-load method calibrates the load cells; ColdCell packs are not tested inside the chamber in this version (D10)	Re-use across the portfolio without adding a battery hazard	ALR-PRC-001 shared blocks
D14	Publish as an open engineering reference with a notice that it is not certified pressure or test equipment; before any use a qualified engineer reviews the vessel calculation and the first pump-down is a proof test behind a guard	Conservative; follows the safety notes in the scaffold	README and every document; build plan safety stops

■ Consequences

- R11 and R12 are added to ALR-REQ-001.
- The vessel, welding, motor supply and safety controls take the estimated cost well above the USD 1,000 value-engineering target (ALR-DEC-001, Value engineering). Amish accepted cost overruns in the same instruction; R10 is posed to Amish with options in `docs/REVIEW.md`.